

Executive Page

Readmissions

Cost Drivers

Satisfaction Insights

Patient Drill Down

Executive Summary

37.1% of patients are at high risk for readmission, primarily due to **cardiac conditions**

Surgery + chemo and cardiac catheterization drive costs above **\$15K per admission**

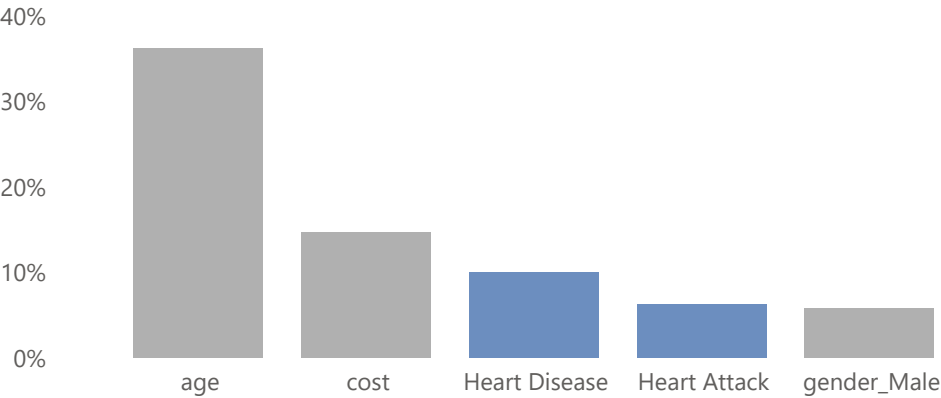
Cardiac patients (2.0-3.5) report lowest satisfaction; **allergic reactions** highest (5.0)

High Risk Readmission %
37.1%

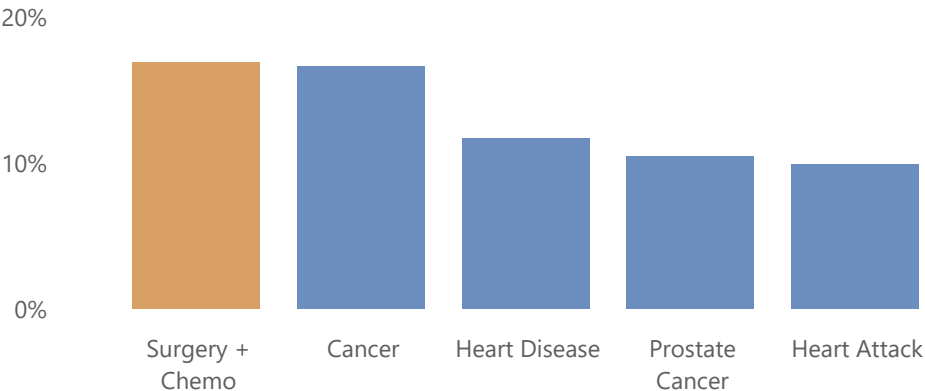
Avg. Predicted Cost
\$8,731

Avg. Predicted Satisfaction
3.6

Top 5 Readmission Causes



Top 5 Cost Drivers



Target	Model	Accuracy	Precision	F1 Score	MAE	R2	RMSE	Recall	ROC AUC
Satisfaction	Gradient Boosting				0.04	1.00	0.00		
Cost	Gradient Boosting				260.63	1.00	91,081.88		
Readmission	Gradient Boosting	1.00	1.00	1.00				1	1.00
Cost	Linear Regression				0.00	1.00	0.00		
Satisfaction	Linear Regression				0.09	0.97	0.03		
Readmission	Logistic Regression	0.88	0.72	0.84				1	0.96
Cost	Random Forest				0.00	1.00	0.00		
Satisfaction	Random Forest				0.00	1.00	0.00		
Readmission	Random Forest	1.00	1.00	1.00				1	1.00

This page summarizes key model outputs and top drivers of readmission and cost.

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Gender

☐ Female

☐ Male

Age Group

☐ 20-29

☐ 30-39

☐ 40-49

☐ 50-59

☐ 60-69

☐ 70-79

LOS

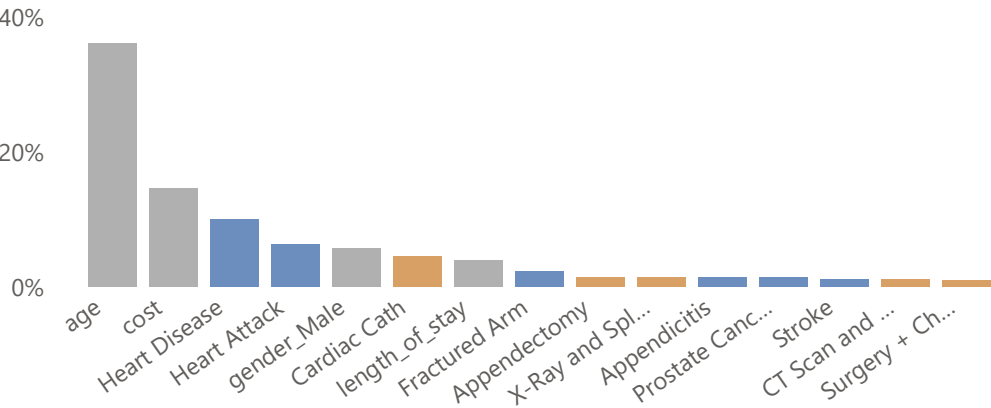
1

76

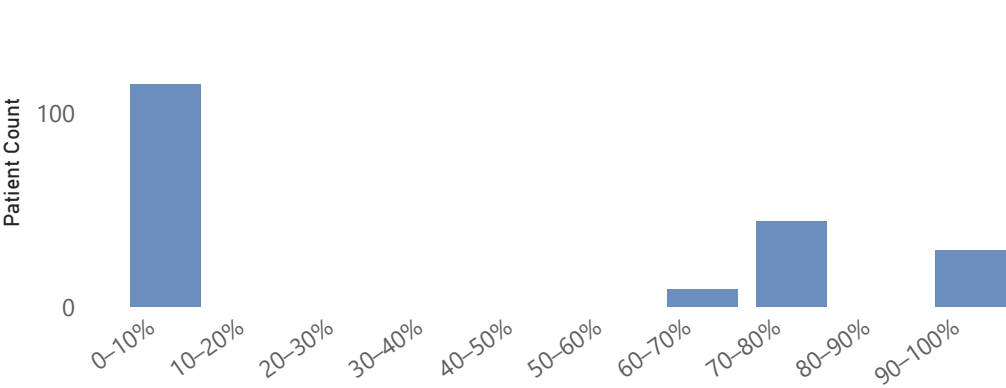


Readmissions Explorer

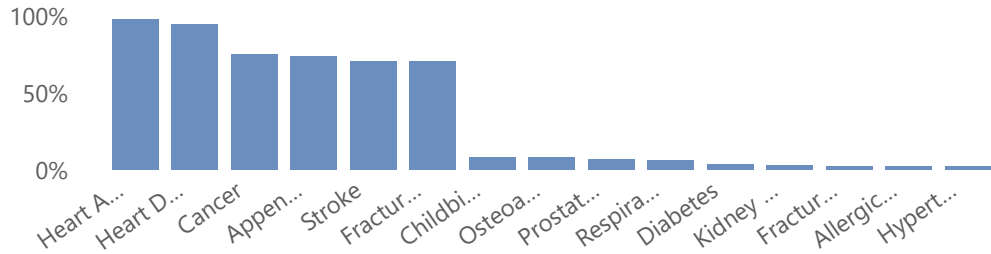
Readmission Feature Importance



Readmission Probability Distribution



Risk by Condition



Risk by Procedure



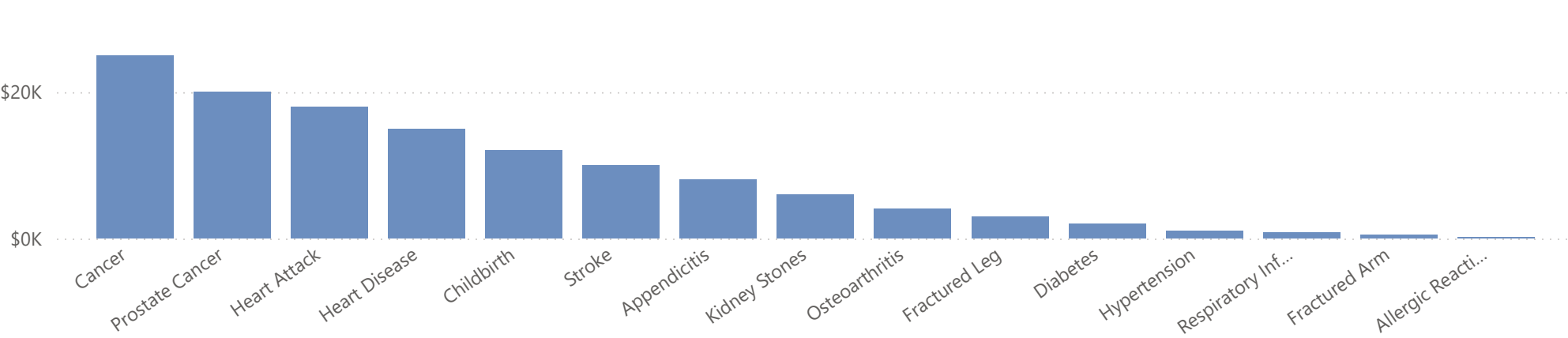
Patient Search

All

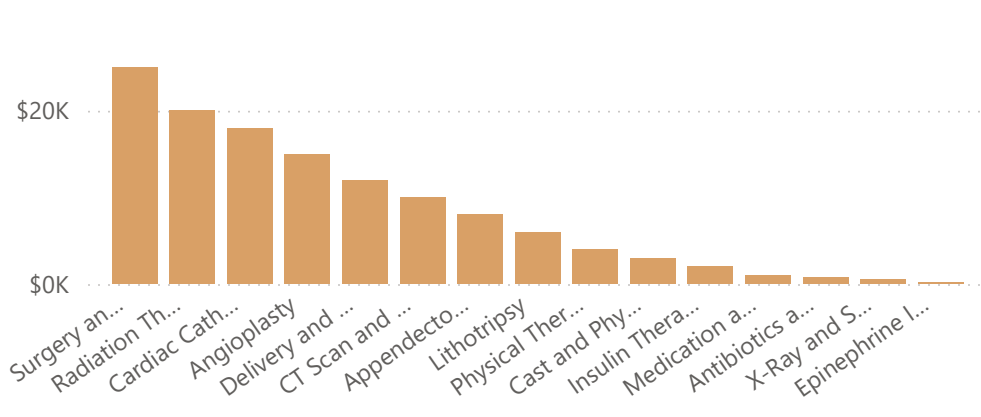
patient_id	condition	Avg Readmission Probability	Avg Predicted Cost	predicted_satisfaction
24	Heart Attack	98.1%	\$18,000	2.02
30	Osteoarthritis	9.0%	\$4,000	4.03
31	Heart Disease	95.4%	\$15,000	3.56
40	Allergic Reaction	2.5%	\$100	5.02
45	Osteoarthritis	9.2%	\$4,000	3.94
50	Cancer	76.7%	\$25,000	4.01
56	Respiratory Infection	7.1%	\$800	3.99

Cost Drivers

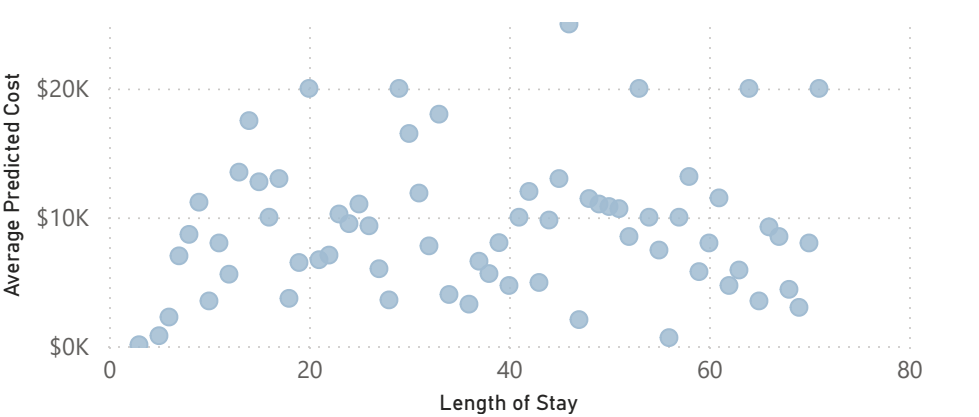
Predicted Cost (\$) by Condition



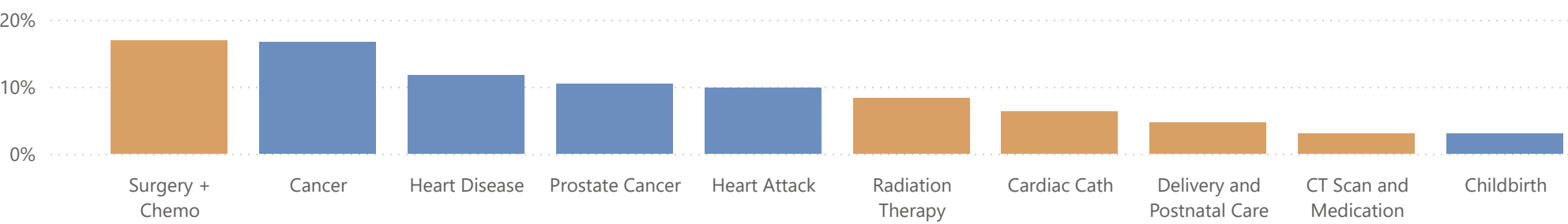
Avg Predicted Cost (\$) by Procedure



Avg Predicted Cost (\$) by Length of Stay (Days)



Top Drivers of Predicted Cost



Age Group

- 20-29
- 30-39
- 40-49
- 50-59
- 60-69
- 70-79

Gender

- Female
- Male

Gender

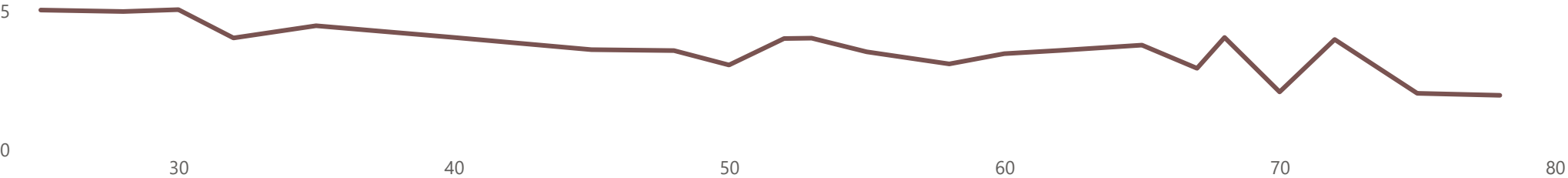
- ☐ Female
- ☐ Male

Age Group

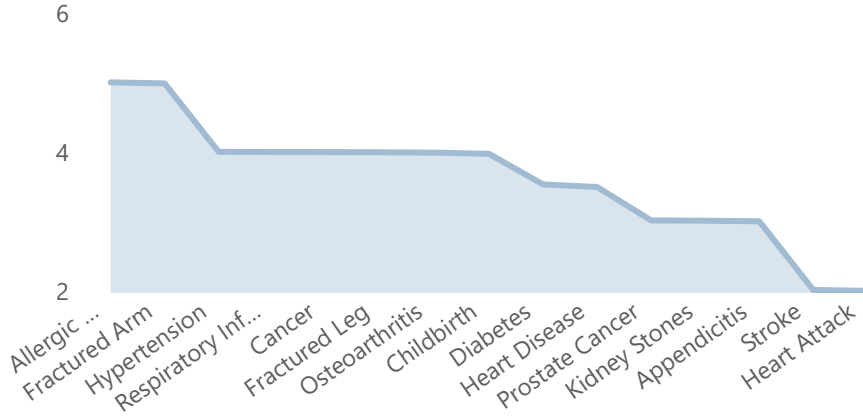
- ☐ 20-29
- ☐ 30-39
- ☐ 40-49
- ☐ 50-59
- ☐ 60-69
- ☐ 70-79

Satisfaction Insights

Predicted Satisfaction by Age



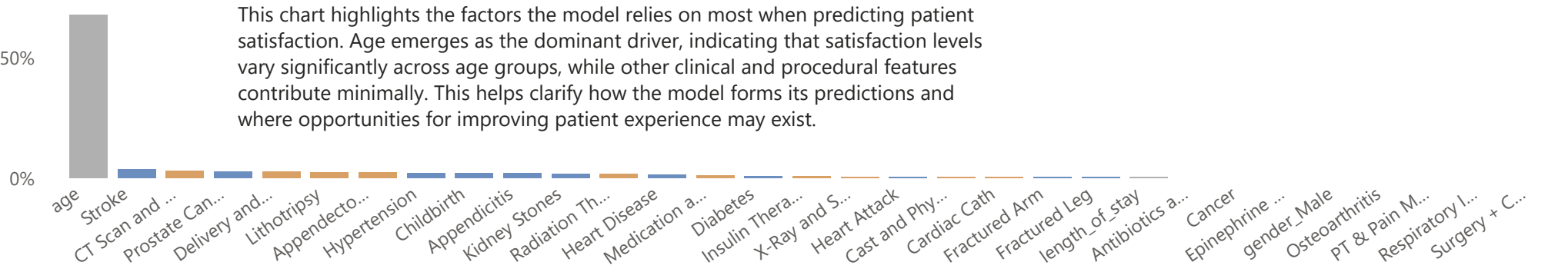
Predicted Satisfaction by Condition



Avg Predicted Satisfaction by LOS



Top Drivers of Predicted Satisfaction



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Patient Drill Down

Patient Drill Down

Age Group

▼

All

▼

Condition

▼

All

▼

Procedure

▼

All

▼

LOS

▼

1

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Satisfaction Score

▼

1

5

Patient ID	Age	Condition	Procedure	LOS	Avg Readmission Probability	Avg Predicted Cost	Avg Predicted Satisfaction
24	75	Heart Attack	Cardiac Catheterization	9	98.1%	\$18,000	2.02
30	60	Osteoarthritis	Physical Therapy and Pain Management	6	9.0%	\$4,000	4.03
31	48	Heart Disease	Angioplasty	7	95.4%	\$15,000	3.56
40	25	Allergic Reaction	Epinephrine Injection	3	2.5%	\$100	5.02
45	65	Osteoarthritis	Physical Therapy and Pain Management	7	9.2%	\$4,000	3.94
50	53	Cancer	Surgery and Chemotherapy	13	76.7%	\$25,000	4.01
56	52	Respiratory Infection	Antibiotics and Rest	5	7.1%	\$800	3.99
61	48	Heart Disease	Angioplasty	9	95.3%	\$15,000	3.56
62	65	Diabetes	Insulin Therapy	7	3.9%	\$2,000	3.51
65	55	Cancer	Surgery and Chemotherapy	14	76.8%	\$25,000	3.97
67	45	Appendicitis	Appendectomy	8	75.7%	\$8,000	3.13
68	40	Fractured Leg	Cast and Physical Therapy	10	2.6%	\$3,000	4.03
69	78	Heart Attack	Cardiac Catheterization	12	98.1%	\$18,000	1.96
72	58	Prostate Cancer	Radiation Therapy	13	7.4%	\$20,000	3.08
74	55	Kidney Stones	Lithotripsy	8	3.2%	\$6,000	2.95
78	35	Fractured Arm	X-Ray and Splint	6	72.9%	\$500	4.95
80	53	Cancer	Surgery and Chemotherapy	15	76.6%	\$25,000	4.01
88	35	Childbirth	Delivery and Postnatal Care	8	9.0%	\$12,000	3.96
90	60	Osteoarthritis	Physical Therapy and Pain Management	10	8.9%	\$4,000	4.03
98	40	Fractured Leg	Cast and Physical Therapy	12	2.5%	\$3,000	4.04
109	78	Stroke	CT Scan and Medication	14	73.0%	\$10,000	1.93